

Visual SLAM — Implementation Progress

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Project Structure

Camera ↓
Feature Matching ↓
Two-View Geometry ↓
Triangulation ↓
PnP / ICP ↓
Nonlinear Optimization (Gauss-Newton / g2o)

Code organization:

include/	public headers
src/	implementations
tests/	one executable per component
reference/slambook/	book source (ch6, ch7)

Camera Model

Pinhole camera, intrinsic matrix K (TUM Freiburg2 calibration used throughout):

$$K = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 520.9 & 0 & 325.1 \\ 0 & 521.0 & 249.7 \\ 0 & 0 & 1 \end{pmatrix}$$

Pixel $\rightarrow$ normalized camera coordinates (`pixel2cam()`):

$$x = K^{-1}p$$

Feature Matching

ORB keypoints + brute-force Hamming matching, filtered by distance $\leq \max(2 \cdot d_{\min}, 30)$.

Quantity	Value
Total matches (before filtering)	500
Good matches (after filtering)	79
Min / Max Hamming distance	4 / 94

All matches



Good matches



Two-View Geometry

Matched points $\Downarrow$
Fundamental Matrix $\Downarrow$
Essential Matrix $\Downarrow$
Recover Pose (R, t)

- ▶ 79 matches used
- ▶ Epipolar constraint $y_2^\top [t]_\times R y_1 \approx 0$ for every match, residuals $\sim 10^{-3}$ — confirms R, t are consistent
- ▶ t is a **unit vector**: monocular two-view geometry has scale ambiguity, direction only

Triangulation

Two-view correspondences $\Downarrow$

Projection matrices $P_1 = K[I|0]$, $P_2 = K[R|t]$ $\Downarrow$

Triangulated 3D points

Points kept only if positive depth in *both* cameras (cheirality check). Depth is in the same unit-norm- t scale as the two-view pose, not metres.



3D points + 2D observations $\Downarrow$
PnP (OpenCV solvePnP) $\Downarrow$
Camera pose $\Downarrow$
Gauss-Newton refinement $\Downarrow$
g2o refinement

- ▶ 75 of 79 matches have valid depth in frame 1 $\rightarrow$ 3D-2D pairs
- ▶ Translation here is **metric** (from real depth, not unit-norm)
- ▶ Cost/ χ^2 convergence (both solvers, same optimum):

40518 $\rightarrow$ 411 $\rightarrow$ 299.76 $\rightarrow$ 299.76

All three methods (solvePnP, Gauss-Newton, g2o) agree to 6 significant figures.

3D-3D correspondences $\Downarrow$
Closed-form SVD (R, t) $\Downarrow$
g2o bundle-adjustment refinement

- ▶ 72 of 79 matches have valid depth in *both* frames
- ▶ Translation is metric (real depth in both point clouds)
- ▶ χ^2 : $1.8161 \rightarrow 1.8155$ — the closed-form SVD solution is already at (or very near) the optimum, so BA has little left to improve
- ▶ Check: $R p_2 + t \approx p_1$ verified on sample points

Current Results

Component	Status
Feature matching (ORB + Hamming)	✓
ORB from scratch (FAST + BRIEF)	✓
Two-view geometry (F/E/H + recoverPose)	✓
Triangulation	✓
PnP	✓
ICP	✓
Gauss-Newton optimization	✓
g2o optimization	✓

All verified against real TUM RGB-D data, numbers reproduced above.

Current Progress

- ▶ Book Chapters 6–7 (*Visual Odometry: Part I*) core algorithms are implemented and tested
- ▶ Feature matching, 2D-2D pose, triangulation, PnP, ICP, and their optimization backends all verified on real data
- ▶ No multi-frame tracking, mapping, or backend yet — single frame-pair components only